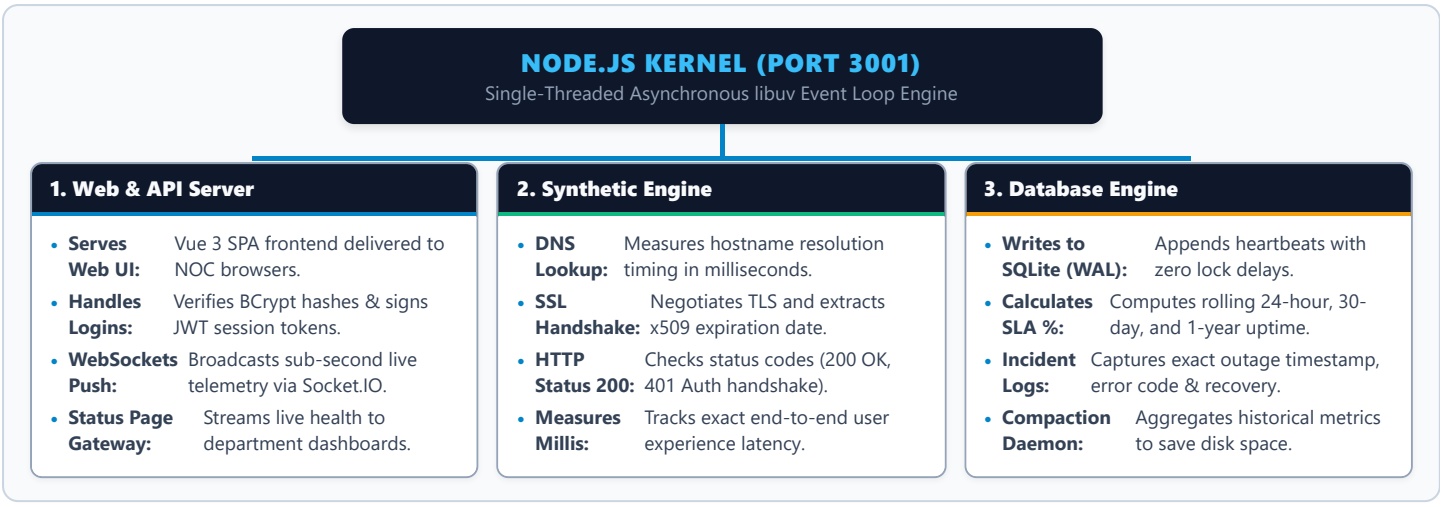
GOVERNMENT OF BIHAR // NATIONAL INFORMATICS CENTRE (NIC)

# Synthetic Probe Execution Timings & Architecture

Duration, Frequency, Latency Milestones, and Network Overhead Blueprint

CURRENT ENGINE
10.133.22.8:3001

**Timing Executive Summary:** Each monitoring cycle operates on two precise time axes: **(1) Execution Duration:** Each individual synthetic transaction takes only **200ms to 450ms** to complete the full DNS, TCP, SSL, and HTTP validation. **(2) Polling Frequency:** Probes trigger every **60 seconds** (1,440 checks/portal/day). If an anomaly occurs, it retries after **60 seconds**, confirming real downtime within **120 seconds** without false alarms.




## 1. CORE TIMING METRICS (LIVE PRODUCTION PARAMETERS)

Production Timers

**PROBE FREQUENCY**
`60 sec`
Standard interval between scheduled synthetic health checks.

**CHECK DURATION**
`240-400 ms`
Time taken to execute each complete L7 synthetic transaction.

**FAULT RETRY TIMER**
`60 sec`
Re-probes once after 60s if the first attempt encounters an error.

**DOWNTIME ALERT**
`≤ 120 sec`
Total time to confirm true outage and dispatch emergency alerts.

## 2. ANATOMY OF A SINGLE PROBE: "HOW LONG DOES 1 PING TAKE?"

Sub-Second Breakdown

| PHASE 1: DNS LOOKUP | PHASE 2: TCP HANDSHAKE | PHASE 3: TLS CRYPTOGRAPHY | PHASE 4: HTTP REQUEST | TOTAL COMPLETED |
|---|---|---|---|---|
| `10 - 35 ms` | `15 - 45 ms` | `70 - 140 ms` | `120 - 220 ms` | `240 - 440 ms` |
| Queries local NOC DNS cache to resolve portal domain name. | Establishes 3-way SYN-ACK handshake with remote web server. | Negotiates SSL cipher and inspects certificate expiration date. | Transmits GET request, receives HTTP 200/401, parses payload. | Complete end-to-end transaction written to SQLite database. |

LIVE FLEET TELEMETRY BENCHMARKS

# Production Portal Latency & Network Overhead Analysis

Measured Telemetry from 10.133.22.8 Across Departmental Endpoints

TELEMETRY SOURCE
LIVE KUMA.DB

## 3. LIVE TELEMETRY BENCHMARKS (RECORDED DIRECTLY ON 10.133.22.8)

Actual Fleet Data

| DEPARTMENT / PORTAL | PROTOCOL | CHECK FREQUENCY | MEASURED LATENCY | CHECKS / 24H | OPERATIONAL STATUS |
|---|---|---|---|---|---|
| **VLTS Transport Portal** (Transport Dept) | HTTPS (IIS Auth Handshake) | Every **60 seconds** | **241 ms – 334 ms** | 1,440 checks | HEALTHY (100% SLA) |
| **ISTP Bihar** (Agriculture Dept) | HTTPS GET (200 OK) | Every **60 seconds** | **315 ms – 434 ms** | 1,440 checks | HEALTHY (100% SLA) |
| **Khanan Soft Bihar** (Mines & Geology) | HTTPS GET (200 OK) | Every **60 seconds** | **347 ms – 396 ms** | 1,440 checks | HEALTHY (100% SLA) |
| **Bihar State NOC Dashboard** (Internal Core) | HTTP GET (LAN Routing) | Every **60 seconds** | **2 ms – 8 ms** | 1,440 checks | LOCAL LOOPBACK |

## 4. NETWORK BANDWIDTH & SERVER OVERHEAD FOOTPRINT

Zero Strain Verification

### NETWORK BANDWIDTH USAGE

- **Payload per probe:** ~2.5 Kilobytes (HTTP HEAD/GET request + headers).
- **Bandwidth per target per minute:** ~0.04 KB/second.
- **Total monthly bandwidth per portal:** ~108 Megabytes.
- **Executive Conclusion:** Negligible network overhead; uses less than **0.001%** of NOC link capacity.

### SERVER HARDWARE RESOURCE IMPACT

- **CPU Consumption:** < 0.5% average CPU load on host `10.133.22.8`.
- **RAM Footprint:** Enforced 512 MB ceiling; actively runs at **~165 MB**.
- **Database Growth:** SQLite appends ~45 KB per day per target.
- **Executive Conclusion:** Zero contention with IIS Port 80 web traffic.

## 5. FLAPPING & OUTAGE ESCALATION LIFECYCLE

Timeline to Alert

| ELAPSED TIME | INTERNAL STATE | SYSTEM ACTION | NOC STATUS |
|---|---|---|---|
| **T = 0s** | Target Fails Probe | Records socket timeout / HTTP 500. Sets state to `PENDING`. | PENDING (NO ALARM) |
| **T = 60s** | Retry Probe Triggered | Initiates second probe attempt to confirm failure. | RE-TESTING |
| **T = 60.3s** | Failure Confirmed | Failure verified twice. State transitions to `DOWN (0)`. | INCIDENT DECLARED |
| **T = 61s** | Incident Dispatched | Turns RED on NOC video wall; dispatches WhatsApp/Telegram alert. | NOC ALERT TRANSMITTED |

**Manager Summary Statement:** *"Our probes run every 60 seconds and complete in roughly 300 milliseconds. If any portal experiences a disruption, our system double-checks it after 60 seconds and alerts the duty officer within 2 minutes total—eliminating false alarms while ensuring instant reaction to real outages."*